Memo version 2.0

BESIII Analysis Memo

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BESIII Analysis Memo (February 4, 2026)

Cross section measurement of process $e^+e^- \to \pi^+\pi^-\pi^0$ from threshold to 2.0 GeV via ISR technique

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Abstract

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List of changes

For an easy reading of the document, major changes in various versions with respect to the previous one are listed below.

Version 1.0

- First version of memo.

Contents

1 Introduction

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2 BEPCII and BESIII detector

The Beijing Electron-Positron Collider II (BEPCII) —a double-ring multi-bunch collider —is operating in the τ-charm energy region (2 to 4.9 GeV) with the design luminosity of 1×10^{33} cm^{-2}s^{-1} [1]. The Beijing Spectrometer III (BESIII) operating at BEPCII is a high-precision, general-purpose detector that records symmetric e^+e^- collisions provided by the BEPCII storage ring. BESIII is approximately cylindrically symmetric with 93% coverage of the solid angle around the e^+e^- interaction point (IP). The components of the apparatus ordered by distance from the IP, are a 43-layer small-cell main drift chamber (MDC), a time-of-flight (TOF) system based on plastic scintillators with two layers in the barrel region and one layer in the end-cap region (the end-cap TOF system has been updated with the multi-gap resistive plate chamber in 2014 [2]), a 6240-cell CsI(Tl) crystal electromagnetic calorimeter (EMC), a superconducting solenoid magnet providing a 1.0 T magnetic field aligned with the beam axis, and resistive plate muon-counter layers interleaved with steel. The momentum resolution for charged tracks in the MDC is 0.5% for transverse momenta with 1 GeV/c. The energy resolution in the EMC is 2.5% in the barrel region and 5.0% in the end-cap region for 1 GeV photons. For charged tracks, particle identification (PID) for charged tracks combines measurements of the energy deposit dE/dx in MDC and flight time in TOF and forms likelihoods $\mathcal{L}(h)(h = K, \pi)$ for a hadron h hypothesis. More details about the BESIII detector are provided in Ref [3].

Table 1: Beam energies and integrated luminosities of the data samples used in this work, quoted from Ref. [4, 5].

E_{beam}(GeV)	Integrated luminosity (pb^{-1})	Data taking time	RunNo
3.773	2931.8 ± 0.2 ± 13.8 [4]	2010(round03) 2011(round04)	11414 – 13988, 14395 – 14604 20448 – 23454
3.773	4995.0 ± 19.0 [5]	2022(round15)	70522 – 73929
3.773	8157.0 ± 31.0 [5]	2023(round16)	74031 – 78536
3.773	4191.0 ± 16.0 [5]	2024(round17)	78615 – 81094

Table 2: Processes and luminosity scales of the inclusive MC samples used in this analysis. Separate samples are available for each data-taking round. Most processes use samples with an integrated luminosity equivalent to ten times that of the data.

Process	Luminosity scale
$e^+e^- \to \psi(3770) \to D^0\bar{D}^0$	10 ×
$e^+e^- \to \psi(3770) \to D^+D^-$	10 ×
$e^+e^- \to \tau^+\tau^-$	10 ×
$e^+e^- \to \psi(3770) \to$ non-$D\bar{D}$	10 ×
$e^+e^- \to q\bar{q}$	10 ×
$e^+e^- \to \gamma_{\text{ISR}}\psi(2S)$	10 ×
$e^+e^- \to \gamma_{\text{ISR}}J/\psi$	10 ×
$e^+e^- \to e^+e^-$	0.25 ×
$e^+e^- \to \gamma\gamma$	3 ×
$e^+e^- \to \mu^+\mu^-$	15 ×

3 Data and MC samples

We use an e^+e^- collision sample corresponding to an integrated luminosity of 20 fb^{-1}, collected by BESIII detector at center-of-mass energy of 3.773 GeV, under BOSS-7.1.2 version. The details are listed in Table 1, quoted from Ref. [4, 5].

Simulated Monte Carlo (MC) samples are employed to study backgrounds, optimize event selection, and estimate detection efficiencies. Two categories of MC samples are utilized in this analysis:

- **The inclusive MC samples** comprehensively model known physics processes at $\sqrt{s} = 3.773$ GeV, including $D\bar{D}$ production, $\tau^+\tau^-$ pair production, initial-state-radiation (ISR) production of vector charmonium states (J/ψ and $\psi(2S)$), QED processes, and continuum light hadron production. The charm working group provides separate samples for the different data-taking rounds(round03-04, round15, round16, and round17) [7]. In this analysis, we use inclusive MC samples with an integrated luminosity equivalent to ten times that of the data for most processes, except for the QED processes $e^+e^- \to e^+e^-$, $\gamma\gamma$, and $\mu^+\mu^-$, where we use the samples with the scaling factors provided by the charm working group. The processes and the corresponding luminosity scales of the samples used in this analysis are summarized in Table 2.

- **Exclusive signal and background MC samples** generated specifically for this analysis using the PHOKHARA event generator [8, 9, 10] (version 9.1). These samples are produced independently for each data-taking round (round03, round04, round15, round16, round17) to account for variations in detector conditions. Sufficient statistics are generated for each process to ensure reliable efficiency determination and background studies. The generated processes include:
  - $e^+e^- \to \pi^+\pi^-\pi^0\gamma_{\mathrm{ISR}}$ (signal process)
  - $e^+e^- \to \pi^+\pi^-\pi^0\pi^0$
  - $e^+e^- \to \pi^+\pi^-\pi^0\pi^0\gamma_{\mathrm{ISR}}$

  These exclusive samples are used for signal efficiency studies, background characterization, and validation of the analysis methodology.

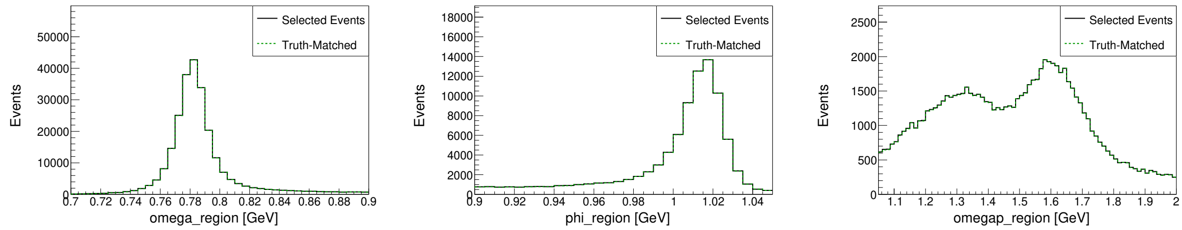
4 Event selections

The process under investigation is $e^+e^- \to \pi^+\pi^-\pi^0\gamma_{\rm ISR}$, whose final states include two charged tracks, one neutral π^0, and one ISR photon. The following selection criteria is applied to all the data samples. Here the figures are shown by taking $\sqrt{s} = 3.773$GeV round16 data as an example. The distributions related to the event selection of other data sets are shown in Appendix. XXX. Situations are similar to each other.

- Good charged track: A charged track is taken as a good MDC track if it originates from the beam vertex and lies within the MDC polar angle region:
  - $|\cos\theta| < 0.93$, where θ is the polar angle with respect to the beam axis.
  - $|\Delta r| < 1$ cm, $|\Delta z| < 10$ cm, where $|\Delta r|$ and $|\Delta z|$ is distance of closest approach to the interaction point (IP) perpendicular to the beam axis and along the beam axis, respectively.
  - the ratio of the calorimeter-deposited energy to the track momentum E_{EMC}/p is required to be less than 0.8 for both of the charged tracks.
  - no PID requirements for π^+, π^-.
- γ selection:
  - $E > 0.025$ GeV (barrel: $|\cos\theta| < 0.8$), $E > 0.050$ GeV (endcap: $0.86 < |\cos\theta| < 0.92$).
  - $0 \leqslant t_{\rm TDC} \leqslant 14(\times 50{\rm ns})$ to suppress fake photons due to electronic noise or beam background.
  - the closest angle with all charged tracks $\theta > 10^\circ$ to suppress background from radiation of charged tracks.
- $\pi^0 \to \gamma\gamma$ selection:
  - two photon invariant mass $M_{\gamma\gamma}$ within (0.115, 0.150) GeV$/c^2$.
  - To improve the momentum resolution, 1C kinematic fit is performed. Here, the 1C means π^0 mass constraint, and the updated 4-momentum will be used in the further analysis. $\chi^2_{\rm KF} < 10$

Candidate events are selected by requiring exactly two good charged tracks, accompanied by at least one reconstructed π^0 and a minimum of three good photon showers. This ensures the presence of at least one π^0 and one isolated photon from Initial-State Radiation ($\gamma_{\rm ISR}$). A four-constraint (4C) kinematic fit is then applied to the event, and the candidate with the smallest $\chi^2_{\rm 4C}$ value is chosen for further analysis.

After identifying an ISR photon from the isolated photon showers, the invariant mass of the remaining $e^+e^- \to \pi^+\pi^-\pi^0$ system is reconstructed. As expected for ISR processes, the resulting center-of-mass energy $\sqrt{s'}$ of the $\pi^+\pi^-\pi^0$ system shows a continuous spectrum, characteristic of the radiative return mechanism. This continuum behavior is illustrated in Fig. 1 across the $\omega(782)$, $\phi(1020)$, and ω'/ω'' mass regions.



(a) $\omega(782)$ region (b) $\phi(1020)$ region (c) ω' and ω'' region

Figure 1: Invariant mass spectra of MC in the (a) $\omega(782)$, (b) $\phi(1020)$, and (c) ω'/ω'' mass regions. The distributions correspond to $M(pK^0_S)$, $M(p\pi^0)$, and $M(K^0_S\pi^0)$, respectively. The "Truth-Matched" distribution (green dashed line) corresponds to reconstructed events where the momentum directions of the π^0 and $\gamma_{\rm ISR}$ successfully match their truth-level counterparts. A loose requirement of $\chi^2_{\rm KF} < 100$ is applied to the kinematic fit. The continuous spectrum of $\sqrt{s'}$ for the $\pi^+\pi^-\pi^0$ system demonstrates the radiative return mechanism via ISR.

5 Background Study

In the study of background processes, the main backgrounds are categorized into $e^+e^- \to \pi^+\pi^-\pi^0\pi^0$, $e^+e^- \to \pi^+\pi^-\pi^0\pi^0\gamma_{\mathrm{ISR}}$, QED processes, and others. The dominant contributions originate from processes with additional π^0s, specifically $e^+e^- \to \pi^+\pi^-\pi^0\pi^0$ and $e^+e^- \to \pi^+\pi^-\pi^0\pi^0\gamma_{\mathrm{ISR}}$. In the inclusive analysis, the cross sections used for these two particular processes were not correctly configured.

To verify that these processes constitute the primary backgrounds, the $\mathcal{L} \times \sigma_{\mathrm{hadron}}$ method is employed, where $\mathcal{L}$ is the integrated luminosity and σ_{hadron} is the hadronic cross section. The luminosity values for each data-taking period are listed in Table 1. The background yield is estimated as a function of the $\pi^+\pi^-\pi^0$ invariant mass (m) using the effective luminosity spectrum, which is related to the mass spectrum via:

$$\frac{dL_{\mathrm{eff}}}{dm} = \frac{2m}{s} \cdot F(s,m) \cdot \mathcal{L}_{\mathrm{int}}, \tag{1}$$

where $s = 4E^2$, E is the beam energy, $\mathcal{L}_{\mathrm{int}}$ is the total integrated luminosity, and $F(s,m)$ is the probability function for initial-state photon emission. This function can be written as [11]:

$$\beta = \frac{2\alpha}{\pi}\left[\ln\left(\frac{s}{m_e^2}\right) - 1\right], \tag{2}$$

$$x = 1 - \frac{m^2}{s}, \tag{3}$$

$$\begin{aligned} \frac{dL_{\mathrm{eff}}}{dm} = \frac{2m}{s} \cdot \Bigg\{ & \beta x^{\beta-1}\left[1 + \frac{\alpha}{\pi}\left(\frac{\pi^2}{3} - \frac{1}{2}\right) + \frac{3}{4}\beta \right.\\ & \left. + \beta^2\left(\frac{37}{96} - \frac{\pi^2}{12} - \frac{1}{72}\ln\frac{s}{m_e^2}\right)\right] \\ & - \beta\left(1 - \frac{x}{2}\right)\ln\frac{1}{x} - \frac{1 + 3(1-x)^2}{x}\ln(1-x) - 6 + x \Bigg\} \cdot \mathcal{L}_{\mathrm{int}}, \end{aligned} \tag{4}$$

where $\alpha = 1/137$ is the fine-structure constant, $m_e = 0.5110034 \times 10^{-3}$ GeV$/c^2$ is the electron mass, and $\sqrt{s} = 3.773$ GeV is the center-of-mass energy of the primary e^+e^- system.

The effective luminosity spectrum calculated using this formula for round16 is shown in Fig. 2. Similar calculations were performed for all data-taking rounds.

Based on this method, rough yield estimates for the two dominant background processes, $e^+e^- \to \pi^+\pi^-\pi^0\pi^0$ (non-ISR) and $e^+e^- \to \pi^+\pi^-\pi^0\pi^0\gamma_{\mathrm{ISR}}$ (ISR), were calculated for the four data-taking periods (round03/04, round15, round16, round17). The results are summarized in Table 3. For the ISR channel, the estimation relies on the $e^+e^- \to \pi^+\pi^-\pi^0\pi^0$ cross section measured by the BaBar collaboration [12].

It is important to note that the values listed in the table are initial yield estimates without any selection efficiency corrections; the actual background contribution to the signal region is obtained by multiplying these yields by the corresponding selection efficiency relevant to the specific analysis cuts. These estimates serve only to confirm that these processes constitute the main sources of background and to provide guidance for optimizing subsequent selection criteria.

The final background yield is then obtained by multiplying the effective luminosity spectrum by the cross section and a selection-efficiency factor that varies dynamically with the analysis cuts. This

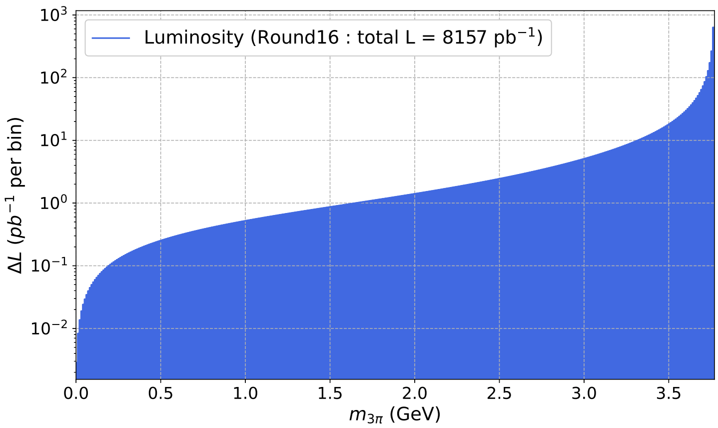


Figure 2: Effective luminosity spectrum as a function of the $\pi^+\pi^-\pi^0$ invariant mass ($m_{3\pi}$) calculated for round16 with total integrated luminosity $\mathcal{L}_{\text{int}} = 8157\ \text{pb}^{-1}$. The spectrum is shown with logarithmic scale on the vertical axis.

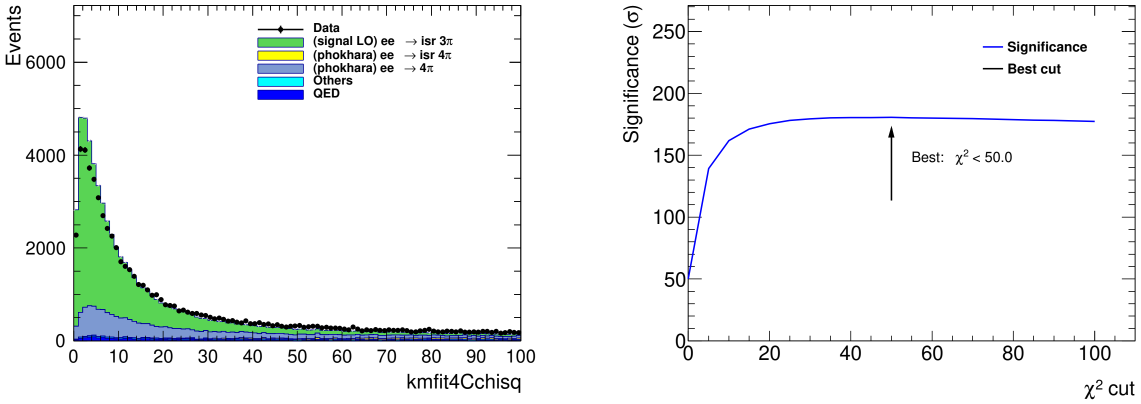
Table 3: Rough estimates of the expected yields for the two dominant background processes before applying any selection efficiency. The integrated luminosities are taken from Table 1. The cross section for $e^+e^- \to \pi^+\pi^-\pi^0\pi^0$ is taken as $\sigma = 0.22$ nb at $\sqrt{s} = 3.773$ GeV, while the energy-dependent cross section for the ISR process is derived from the measured $e^+e^- \to \pi^+\pi^-\pi^0\pi^0$ cross section [12]. The final background contribution is obtained by multiplying these yields by the corresponding selection efficiency.

Data sample (Run period)	$N(e^+e^- \to \pi^+\pi^-\pi^0\pi^0)$	$N(e^+e^- \to \pi^+\pi^-\pi^0\pi^0\gamma_{\text{ISR}})$
round03-04	511940	1142096
round15	872209	1945825
round16	1424346	3177597
round17	731817	1632623

method is used solely to confirm that $e^+e^- \to \pi^+\pi^-\pi^0\pi^0$ and $e^+e^- \to \pi^+\pi^-\pi^0\pi^0\gamma_{\text{ISR}}$ are the dominant backgrounds and to optimize the selection criteria. It is not used to derive the final background estimates; precise yields for these two components will be determined using data-driven methods.

Based on the rough background estimate from $\mathcal{L} \times \sigma_{\text{hadron}}$, the signal yield was approximated by subtracting the background from the data, allowing for a preliminary significance calculation. Using this approach, a figure of merit (FOM) defined as $S/\sqrt{S+B}$ was constructed to evaluate the signal significance. To optimize the event selection, a scan over the kinematic fit χ^2 was performed, with the result shown in Fig. 3. The Round 16 optimization yields an optimal cut of $\chi^2_{\text{KF}} < 50$.

Based on the optimal χ^2_{KF} selection, the invariant mass spectrum of the 3π system ($M_{3\pi}$) can be divided into three distinct regions for further study: the $\omega(782)$ region, the $\phi(1020)$ region, and a higher-mass region associated with possible ω'/ω'' excitations. In the high-mass ω'/ω'' region, a significant background component from events containing additional π^0 mesons was observed. To quantify this, the number of π^0 candidates per event was counted, where any π^0 candidate satisfying a one-constraint



(a) χ^2 distribution for the 4C kinematic fit

(b) FOM scan as a function of the χ^2 cut

Figure 3: (a) Distribution of the kinematic fit χ^2 for the four-constraint (4C) fit using the Round 16 data sample. (b) The corresponding figure of merit ($S/\sqrt{S+B}$) as a function of the applied χ^2 cut value for the Round 16 data.

kinematic fit $\chi^2 < 200$ was accepted. The distribution of this π^0 multiplicity is shown in Fig. 4. To suppress this specific background, a requirement of exactly one π^0 candidate per event was applied exclusively in the ω'/ω'' mass region. No such restriction was applied to the $\omega(782)$ and $\phi(1020)$ signal regions. The following studies and distributions are based on the Round 16 data sample.

The resulting $M_{3\pi}$ invariant mass spectra, after applying all selection criteria including the region-specific π^0 multiplicity cut, are presented separately for the three mass regions in Fig. 5.

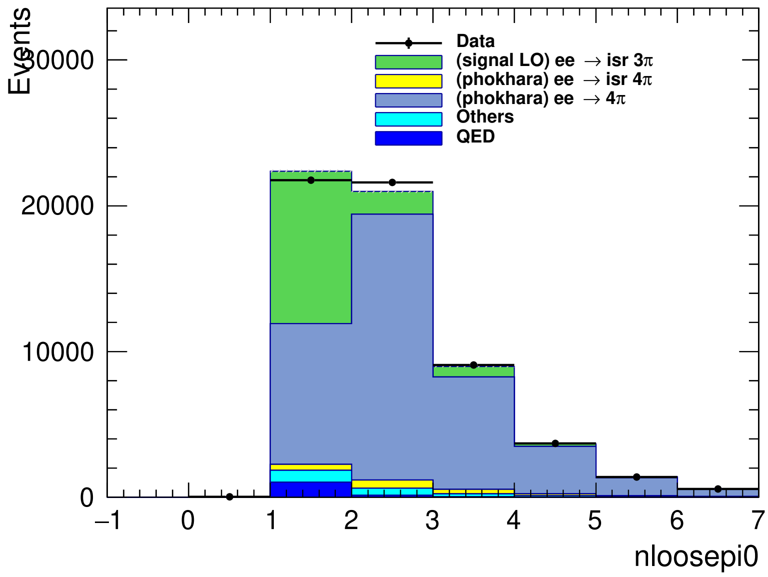
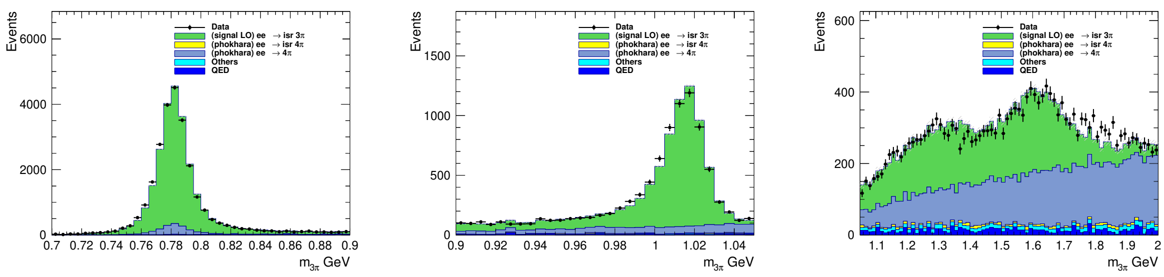


Figure 4: Distribution of the number of π^0 candidates per event in the ω'/ω'' mass region using the Round 16 data. A π^0 candidate is counted if its 1C kinematic fit $\chi^2 < 200$.



(a) $\omega(782)$ region

(b) $\phi(1020)$ region

(c) ω'/ω'' region

Figure 5: Invariant mass spectra $M_{3\pi}$ for the (a) $\omega(782)$, (b) $\phi(1020)$, and (c) ω'/ω'' mass regions obtained from the Round 16 data. The selection for the high-mass region (c) includes the requirement $N_{\pi^0} = 1$, while regions (a) and (b) have no π^0 multiplicity cut.

5.1 estimate of $e^+e^- \to \pi^+\pi^-\pi^0\pi^0$

In estimating the $e^+e^- \to \pi^+\pi^-\pi^0\pi^0$ background, the following strategy was adopted:

It is assumed that the total yield of $e^+e^- \to \pi^+\pi^-\pi^0\pi^0$ events produced in the data is a fixed quantity. This process enters our signal region primarily when a photon from one of the π^0 decays is misidentified as an initial-state radiation photon ($\gamma_{\rm ISR}$). Consequently, two distinct selection efficiencies can be calculated: $\epsilon_{\rm sig}$ (the efficiency for $e^+e^- \to \pi^+\pi^-\pi^0\pi^0$ events to be selected as signal candidates) and $\epsilon_{\rm bkg}$ (the efficiency for the same process to enter the 3π signal region as background).

When $e^+e^- \to \pi^+\pi^-\pi^0\pi^0$ events are selected as signal, the same analysis procedure as for the signal is applied to construct the 3π invariant mass spectrum, thereby obtaining the distribution of this process in data. Using the ratio of these two efficiencies, $R = \epsilon_{\rm bkg}/\epsilon_{\rm sig}$, the observed $e^+e^- \to \pi^+\pi^-\pi^0\pi^0$ signal distribution can be converted into an estimate of its background distribution in the signal region. Specifically:

$$N_{\rm bkg} = R \times N^{\rm obs}_{\rm sig} = \frac{\epsilon_{\rm bkg}}{\epsilon_{\rm sig}} \times N^{\rm obs}_{\rm sig},$$

where $N^{\rm obs}_{\rm sig}$ is the number of observed $e^+e^- \to \pi^+\pi^-\pi^0\pi^0$ signal events in data, and $N_{\rm bkg}$ is the estimated number of background events from this process in the signal region.

This method provides a data-driven conversion from the observed signal distribution to the background distribution through efficiency correction.

When treating $e^+e^- \to \pi^+\pi^-\pi^0\pi^0$ as the signal, the selection criteria are designed such that the same event sample also satisfies a loose selection for $e^+e^- \to \gamma_{\rm ISR}\pi^+\pi^-\pi^0$. This ensures that the invariant mass of the three-pion system is automatically defined within the latter process.

The combined selection proceeds as follows: events are required to meet the criteria for both processes. The first step applies the full $e^+e^- \to \pi^+\pi^-\pi^0\pi^0$ selection:

- **$\pi^+\pi^-\pi^0\pi^0$ Candidate Selection:**
  - Exactly two good charged tracks.
  - nPi0 ≥ 2 & & nGoodShowers ≥ 4 (ensuring at least two π^0s).
  - Application of a total 4-momentum constraint to the initial e^+e^- collision (4C kinematic fit).
  - The candidate with the minimum $\chi^2_{\rm 4C}$ is chosen.
  - A requirement of $\chi^2_{\rm KF} < 10$ on the kinematic fit.

Subsequently, within the same event, a candidate for $e^+e^- \to \gamma_{\rm ISR}\pi^+\pi^-\pi^0$ is formed by reinterpreting one of the π^0s as the $\gamma_{\rm ISR}$:

- **$\gamma_{\rm ISR}\pi^+\pi^-\pi^0$ Candidate Selection:**
  - The same two good charged tracks.
  - nPi0 ≥ 1 & & nGoodShowers ≥ 3 (ensuring at least one π^0 and one photon assigned as $\gamma_{\rm ISR}$).
  - Application of the same total 4-momentum constraint to the initial e^+e^- collision (4C kinematic fit).
  - The candidate with the minimum $\chi^2_{\rm 4C}$ is chosen.

– No additional kinematic fit χ^2 requirement is applied at this stage.

This dual-tag strategy guarantees that the reconstructed 3π system from the $e^+e^- \to \gamma_{\mathrm{ISR}}\pi^+\pi^-\pi^0$ hypothesis is kinematically well-defined, and its invariant mass can be reliably calculated for the subsequent analysis.

The selection described above results in a highly pure sample of $e^+e^- \to \pi^+\pi^-\pi^0\pi^0$ events. The invariant mass spectrum of the corresponding three-pion system from these events, which directly enters the background estimation, is shown in Fig. 6. This observed distribution in data will be scaled by the efficiency ratio $R = \epsilon_{\mathrm{bkg}}/\epsilon_{\mathrm{sig}}$ to estimate the final background contribution in the signal region. All other backgrounds will be subtracted directly.

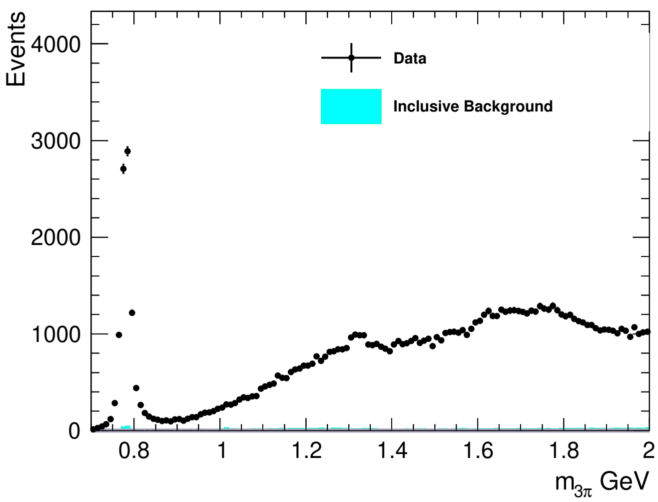


Figure 6: The 3π invariant mass spectrum from selected $e^+e^- \to \pi^+\pi^-\pi^0\pi^0$ events (red points), overlaid with the scaled Monte Carlo simulation (blue histogram). The distribution is obtained after applying the dual-tag selection criteria and represents the pure $e^+e^- \to \pi^+\pi^-\pi^0\pi^0$ component in data.

5.2 estimate of $e^+e^- \to \pi^+\pi^-\pi^0\pi^0\gamma_{\text{ISR}}$

The second dominant background, $e^+e^- \to \gamma_{\text{ISR}}\pi^+\pi^-\pi^0\pi^0$, is estimated using an analogous data-driven approach. The selection follows a similar dual-tag strategy to obtain a sample. The selection criteria are outlined below:

Simultaneously, events must satisfy both the $e^+e^- \to \gamma_{\text{ISR}}\pi^+\pi^-\pi^0\pi^0$ selection, where the γ_{ISR} is treated as an untagged missing momentum, and the tagged $e^+e^- \to \gamma_{\text{ISR}}\pi^+\pi^-\pi^0$ loose selection criteria.

- $e^+e^- \to \gamma_{\text{ISR}}\pi^+\pi^-\pi^0\pi^0$ **Candidate Selection (Untagged γ_{ISR}):**
  - Exactly two good charged tracks.
  - nPi0 ≥ 2 && nGoodShowers ≥ 4 (requiring at least two π^0s).
  - The γ_{ISR} is not explicitly reconstructed from a shower. Instead, its 4-momentum is inferred as a *missing* component within the kinematic fit.
  - A total 4-momentum constraint to the initial e^+e^- collision is applied (4C fit).
  - The candidate with the minimum $\chi^2_{4\text{C}}$ is chosen.
  - A requirement of $|\cos\theta_{\text{Miss}}| > 0.99$ is applied to the direction of the missing 4-momentum to ensure it is consistent with an ISR photon along the beam axis.
  - A kinematic fit quality cut of $\chi^2_{\text{KF}} < 10$ is applied.
- $\gamma_{\text{ISR}}\pi^+\pi^-\pi^0$ **Candidate Selection (within the same event):**
  - The same two good charged tracks.
  - nPi0 ≥ 1 & & nGoodShowers ≥ 3 (ensuring at least one π^0 and one photon tagged as γ_{ISR}).
  - A total 4-momentum constraint to the initial e^+e^- collision is applied (4C fit).
  - The candidate with the minimum $\chi^2_{4\text{C}}$ is chosen.
  - No additional kinematic fit χ^2 requirement is applied.

The application of this dual-tag selection strategy resulted in a sample with minimal contamination from other processes. The high purity observed is a consequence of the stringent criteria. The corresponding 3π invariant mass distribution from this sample, which will be used for the background estimation, is shown in Fig. 7.

The observed distribution will be similarly scaled by the appropriate efficiency ratio $R' = \epsilon'_{\text{bkg}}/\epsilon'_{\text{sig}}$ to estimate its contribution in the signal region.

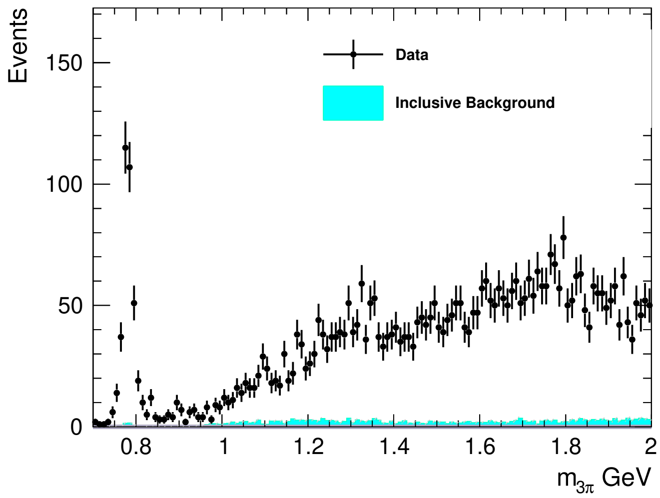


Figure 7: The 3π invariant mass spectrum from selected $e^+e^- \to \gamma_{\rm ISR}\pi^+\pi^-\pi^0\pi^0$ events (green points). The distribution is obtained after applying the dual-tag selection and serves as a clean template for background estimation.

6 Unfolding

to fill

7 Efficiency correction

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8 Systematic uncertainties

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9 Summary

to fill

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APPENDIX